

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree Supplementary Examination May 2023 (2019 Scheme)

Course Code: CST302**Course Name: COMPILER DESIGN**

Max. Marks: 100

Duration: 3 Hours

PART A**Answer all questions, each carries 3 marks.**

Marks

- 1 Construct a regular expression for the language that consists of all strings ending with 00 over $\Sigma = \{0,1\}$. (3)
- 2 Define tokens, lexemes and patterns with suitable examples for each. (3)
- 3 Given a grammar : (3)
$$S \rightarrow (L) | a$$
$$L \rightarrow L, S | S$$
 - (i) Is the grammar ambiguous? Justify.
 - (ii) Build a parse tree for the string $(a, ((a, a), (a, a)))$.
- 4 Compute the FIRST and FOLLOW for the following Grammar. (3)
$$S \rightarrow SS | AB$$
$$A \rightarrow Aa | a$$
$$B \rightarrow Bb | b$$
- 5 Define an operator grammar. Give an example. (3)
- 6 What are viable prefixes? (3)
- 7 Explain quadruples, triples and indirect triples with suitable examples. (3)
- 8 What are L-attributed definitions and S-attributed definitions in a syntax directed translation scheme? (3)
- 9 Construct the syntax tree and then draw the DAG for the statement: (3)
$$e := (a*b) + (c-d) *(a*b)$$
- 10 Explain any three issues in the design of a code generator. (3)

PART B**Answer one full question from each module, each carries 14 marks.****Module I**

- 11 a) Explain in detail the various phases of the compiler with a neat diagram. (9)
Illustrate the output of each phase for the input,
$$\text{sum} := a + b * 30$$

where a and b are float variables.

- b) Apply bootstrapping to develop a compiler for a new high level language N on machine P. (5)

OR

- 12 a) Explain the role of transition diagrams in recognition of tokens. Draw the transition diagram for the regular definition: (8)

$$\text{relop} \rightarrow < | <= | = | <> | >= | >$$

- b) List and explain any three tools that help a programmer in building a compiler efficiently. (6)

Module II

- 13 a) Consider the following grammar: (9)

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow \sim F \mid (E) \mid \text{id}$$

- (i) Remove left recursion from the grammar.
 (ii) Construct a predictive parsing table.
 (iii) Justify the statement “ The grammar is LL (1)”.

- b) Design a recursive descent parser for the grammar: $S \rightarrow cAd$, $A \rightarrow ab/b$ (5)

OR

- 14 a) Left factor the following grammar and then obtain LL(1) parsing table. (7)

$$S \rightarrow TL;$$

$$T \rightarrow \text{int} \mid \text{float}$$

$$L \rightarrow L, \text{id} \mid \text{id}$$

Is the grammar LL(1)? Justify your answer.

- b) Write all the moves by the LL(1) parser for parsing the input “int id,id;”. [Use the parsing table created in question number 14.a] (7)

Module III

- 15 a) Construct canonical LR(0) collection of items for the grammar below. (10)

$$S \rightarrow L = R$$

$$S \rightarrow R$$

$$L \rightarrow * R$$

$$L \rightarrow \text{id}$$

$$R \rightarrow L$$

Also identify a shift reduce conflict in the LR(0) collection constructed above.

- b) Write an algorithm for computing the closure of an LR(0) items. (4)

OR

- 16 a) Construct LALR parse table for the grammar: $A \rightarrow BB, B \rightarrow bB \mid d$ (10)
b) What are the different parsing conflicts in the SLR parsing table? (4)

Module IV

- 17 a) Write the SDD for a desk calculator and draw the annotated parse tree for the expression: $4 * 5 + 6 - (3 * 2)$ (8)
b) Explain bottom- up evaluation of s-attributed definitions. (6)

OR

- 18 a) Write syntax directed definition to construct syntax tree and three address code for assignment statements. (7)
b) Explain static allocation and heap allocation strategies. (7)

Module V

- 19 a) With suitable examples explain the following loop optimization techniques: (7)
(i) Code motion (ii) Induction variable elimination and (iii) strength reduction
b) Explain the optimization of basic blocks. (7)

OR

- 20 a) For the following C statement, write the three-address code and quadruples. (8)

$S := A - B + C * D - E + F$

Also convert the three-address code into machine code.

- b) Write the Code Generation Algorithm and explain the *getreg* function. (6)
